

# Speaking Script

*Decentralized Robot Mobile Fulfillment  
System Control with World Models*

|              |                                        |
|--------------|----------------------------------------|
| Audience     | Advisor — research progress update     |
| Total length | 28 min talk + 5 min Q&A                |
| Language     | English                                |
| Slide count  | 24 main slides (1 title + 23 numbered) |
| Style        | Confident · honest · narrative-driven  |

## §1 Hook &amp; Background

5 min · 0:00 – 5:00

## Title slide

0:00 – 0:30 · 30 sec

*Decentralized RMFS with World Models***Goal:** Visual hook. The dream rollout is already looping. Establish identity and title.*[The ground-truth-vs-dream GIF is already looping behind the title before you begin. Stand silent for 3 seconds.]***“Good afternoon, Professor.”***(pause 1 s)***“What you see behind me is not a simulator recording.”***(pause 2 s)***“It is the *imagination of a robot* — a model dreaming the future of a warehouse, frame by frame, without ever running the real simulator.”***(pause 2 s)***“The title: Decentralized Robot Mobile Fulfillment System Control with World Models. I am [Name], advised by Prof. Shou Yan Chou. NTUST · CITI Lab. This is my research progress update.”**

Don't point at the GIF — let it play behind you. Eye contact on “the imagination of a robot.”  
→ Click into Slide 01 confidently.

## Slide 01 · 23

0:30 – 2:00 · 1.5 min

*Why Centralized RMFS is Fragile***Goal:** Identify the enemy — centralized control. Set up the tension that resolves on Slide 03.**“Modern RMFS — Amazon Robotics, Geek+ — runs on centralized control. A single central planner — typically a MAPF solver like CBS, BCP, or A\* — computes paths for the entire fleet simultaneously.”***[point at the diagram on the left — central node with radiating links and the warning indicator]***“Three properties of this architecture. One, a single computer dictates the path of every robot. Two, continuous communication is required — TCP backbone, no graceful degradation. Three, compute grows combinatorially with fleet size — coordination is the bottleneck.”***(pause 2 s)***“Elegant in theory. But the entire operation lives or dies with one process.”**

The diagram has a pulsing red warning indicator at the centre — let it speak. “Lives or dies with one process” — slow, deliberate. Set up the next slide.

## Slide 02 · 23

2:00 – 3:30 · 1.5 min

*Failure Modes***Goal:** Make the fragility concrete. Three icon cards plus an impact table.*“Three failure modes, one shared consequence.”**[point at the three cards across the top]**“**Mode 1 — Central Server Crash.** A single OS-level failure halts the planner. Every robot freezes mid-path. There is no fallback.”**“**Mode 2 — Network Partition.** TCP or wireless dropout strands robots without new paths. They either stop or run a stale plan into a wall.”**“**Mode 3 — Robot Sensor Fault.** A single bot reporting an unknown state triggers full-fleet replanning — combinatorial cost.”**[point at the table below]**“The operational impact: server crash equals 100% downtime. Network partition equals local zone halt. A single robot failure pauses everyone. A picker delay cascades into throughput drop.”**(pause 2 s)**“Every component capable of breaking becomes a vector for system-wide stoppage.”*

Three cards spoken with rhythm. The table is read briskly — the advisor doesn't need every word. → Build into Slide 03.

## Slide 03 · 23

3:30 – 4:30 · 1 min

*Decentralized Vision***Goal:** Resolve the tension. State the three thesis properties.*“What if each robot could decide on its own?”**[point at the diagram on the left — eight robots each with a  $\pi$  brain badge]**“Same warehouse, but no central server. Each robot owns a local policy. One robot fails — the rest keep operating. The challenge: how does a robot make smart decisions without seeing the whole map?”**(pause 2 s)**“The traditional answer is rule-based. The modern answer — what I am working on — is a **world model**: the robot learns from its visual observations and can imagine its future.”**[point at the three property cards on the right]**“Three properties this work commits to. One — Decentralized: no central planner, each robot decides on board. Two — Vision-only: pixels in, action out, no coordinates. Three — Imagination-trained: the controller never touches the real environment during training.”*

“Brain on board” while pointing at one of the bots. Three properties spoken slowly — counting on fingers helps. → “And to understand how this is possible, we first need to define what a world model is.”

## §2 World Models · Definition

4 min · 5:00 – 9:00

Slide 04 · 23

4:30 – 6:00 · 1.5 min

*What Is a World Model?*

Goal: Definition. Use the architecture diagram at the top to anchor the explanation.

*“World models — popularised by David Ha and Jürgen Schmidhuber, 2018. The architecture diagram at the top of this slide is from their paper, redrawn for our setting.”*

[point through the diagram]

*“Environment at the top. Observation on the left. The world model in the dashed box containing the VAE bowtie and the MDN-RNN box. The controller wedge on the right. Arrows in amber for latent  $z$ , violet for hidden state  $h$ , coral for action. The animated tokens that ride along the arrows show the dataflow. We will open up  $V$ ,  $M$ , and  $C$  one slide at a time.”*

[point at the equation]

*“Formally:  $p$  of  $z$ - $t$ -plus-1 given  $z$ - $t$ ,  $a$ - $t$ ,  $h$ - $t$ . Given the current latent state, the action taken, and the hidden history, the model predicts the next latent state.”*

*“Agents can use the world model to plan a sequence of actions in their dreams.”*  
— Ha & Schmidhuber, 2018.

[point at the bullets]

*“Three properties. Trained from observation data alone — no physics engine. Can be rolled out forward to imagine futures with no simulator in the loop. And the policy can be trained in dreams instead of reality — cheap, parallel, safe.”*

Walk through the diagram before reading the equation — visual first. Read the quote slowly. →  
“Now let’s open the world model into its three components.”

Slide 05 · 23

6:00 – 7:30 · 1.5 min

 $V \cdot M \cdot C$ 

Goal: Roadmap for the rest of the talk. Three panels with mini-architecture diagrams.

*“Three components, following an analogy with human cognition.”*

[point at the  $V$  panel]

*“**V — Vision.** Compress pixels to latent. The mini-diagram shows it: a  $96 \times 96$  RGB input image goes into the VAE bowtie — encoder narrows to a 64-dimensional latent  $z$ , decoder mirrors back to a reconstructed image. Implementation: Variational Autoencoder.”*

[point at the  $M$  panel]

*“**M — Imagination.** Predict the next latent and the reward. The diagram: the triple  $(z, a, h)$  enters the MDN-RNN box, output is  $\hat{z}'$  and  $\hat{r}$ . The recurrent loop on top is the LSTM hidden state being fed back. Implementation: MDN-RNN with a 5-component Gaussian mixture head plus a reward head.”*

[point at the *C* panel]

**“*C* — *Decision*. A tiny linear policy. The diagram: the violet arrow  $[z \oplus h]$  enters the *C* wedge, the coral arrow *a* exits to the right. One matrix plus a bias — 320 inputs to 11 outputs — 3,531 parameters.”**

(pause 2 s)

**“The key insight: *V* and *M* are trained from offline data. *C* is trained inside dream rollouts of *V* and *M* — it never touches the real simulator. This is why the world-model approach is so sample-efficient.”**

Use the three mini-diagrams as visual anchors — point directly at each. Pause between *V*, *M*, and *C*. “3,531 parameters” — second mention. Plant it. → Slide 06.

## Slide 06 · 23

7:30 – 8:30 · 1 min

Lineage · Two Tracks

**Goal:** Position the work within state of the art. Two parallel timelines: amber Pixel-Reconstruction track and teal JEPA track. Bottom strip has three panels.

**“World-model research has split into two parallel lineages.”**

[point at the top track]

**“*Track A* — *Pixel Reconstruction*. 2018: Ha and Schmidhuber — the highlighted anchor, our baseline. 2019–22: the Dreamer family — RSSM, gradient actor, master 150 tasks. 2022: DayDreamer and Plan2Explore — real robots and curiosity in dream space. 2024: IRIS and TWM — transformer dynamics with discrete tokens. 2024–25: Sora and DeepMind’s Genie 3 — generative world simulators, real-time interactive. 2026: NVIDIA Cosmos — Predict 2, Reason 2, Cosmos Policy, Alpamayo — world foundation models in production.”**

[point at the bottom track]

**“*Track B* — *Embedding Prediction* · *JEPA*. 2022: LeCun’s JEPA position paper — predict in embedding space, not pixels. 2023: I-JEPA — image self-supervised. 2024: V-JEPA — Meta’s video version, two million unlabelled videos. 2025: V-JEPA 2 with the action-conditioned variant deployed zero-shot on Franka arms. 2026: LeJEPA — the theoretical foundation.”**

[point at the bottom strip]

**“Why H&S 2018 as the base? *Smallest controller* — 3,531 parameters. *CMA-ES* is gradient-free, handles discrete actions cleanly. *JEPA is the orthogonal axis* I want to explore later. The H&S adaptations specific to my work — frame stacking, override-aware action, sparse warehouse reward — are listed on the right.”**

Read each timeline briskly — don’t dwell on any single node. “JEPA is the orthogonal axis” — set up future work. → Slide 07.

### §3 Methodology

5 min · 9:00 – 14:00

Slide 07 · 23

8:30 – 9:30 · 1 min

*Methodology Overview*

Goal: Pipeline as a flow plus four adaptations against H&amp;S.

**“Pipeline follows H&S, with specific adaptations for RMFS.”**

[point at the five-step flow at the top]

**“Five steps: RMFS data  $\rightarrow$  V offline  $\rightarrow$  M offline  $\rightarrow$  C via CMA-ES in dreams  $\rightarrow$  per-robot deployment. Roughly 1 millisecond per decision.”**

[point at the adaptations table below]

**“Four adaptations against H&S 2018. Environment: VizDoom  $64 \times 64 \rightarrow$  RMFS warehouse  $96 \times 96$ . Frame stacking: single frame  $\rightarrow$  2-frame stack for motion cueing. Action space: 3 discrete  $\rightarrow$  11-dim with override, direction, accel and decel. Reward: continuous game score  $\rightarrow$  sparse warehouse reward — throughput minus collisions.”**

The flow + table speak for themselves. Read briskly.  $\rightarrow$  Slide 08.

Slide 08 · 23

9:30 – 11:00 · 1.5 min

*V · VAE Visual Encoder*

Goal: Explain V. The reconstruction at training epoch 99 is on the right.

**“First component, V — the VAE encoder.”**

[point at the architecture on the left]

**“Input is  $96 \times 96 \times 6$  — that’s the 2-frame stack. Four convolutional layers, each stride 2:  $96 \rightarrow 47 \rightarrow 22 \rightarrow 10 \rightarrow 4$ . Then fully connected to  $\mu$  and  $\log \sigma$ , each 64-dimensional. The decoder mirrors back.”**

(pause 2 s)

**“Important design choice: the decoder reconstructs only the 3-channel current frame, not the full 6-channel stack. We want  $z$  to encode current state; history is M’s job.”**

[point at the reconstruction on the right]

**“On the right, the actual reconstruction at training epoch 99 — straight from the checkpoint. Loss: MSE plus  $\beta \cdot \text{KL}$  with  $\beta = 1.0$ . Training time about 2 hours on a consumer GPU.”**

[point at the pills below]

**“The compression:  $96^2 \times 6$  channels  $\rightarrow$  64-d  $z$ . Roughly  $1,300\times$  compression.”**

Layer numbers spoken briskly. “Decoder reconstructs only the current frame” — slow, this is non-trivial.  $\rightarrow$  Slide 09.

## Slide 09 · 23

11:00 – 12:30 · 1.5 min

*M · MDN-RNN Dynamics*

Goal: Explain M. Use the GMM five-futures visualization at the bottom.

*“Second component, M — the MDN-RNN.”**[point at the architecture on the left]**“Input 75-dimensional: 64-d  $z$  plus 11-d action. LSTM hidden 256. Output 646 dimensions, factored into  $\mu_k, \sigma_k, \pi_k$  for  $k = 1..5$ , plus the reward head  $\hat{r}$ .”**[point at the equation panel on the right]**“p of z-t-plus-1 given z-t, a-t, h-t equals the sum over k of  $\pi_k$  times Gaussian  $\mu_k \sigma_k^2$ . Plain English: five possible futures, weighted.”**[point at the bullets on the right]**“Why a GMM? Because warehouse dynamics are multimodal — at a junction, the bot might go left or right; a single Gaussian would average between them. The LSTM hidden state  $h_t$  handles partial observability. And the reward head  $\hat{r}$  — that is what CMA-ES will use as fitness in dreams.”**[point at the visualization at the bottom — full-width media frame]**“And here is what “five possible futures” actually looks like.”**(pause 2 s)**“Left: the seed observation reconstructed by the VAE. Right: five frames, one decoded from each Gaussian component —  $k = 1$  through 5. Below each, the mixture weight  $\pi_k$  with a small bar.”**[point along the bars]**“For this seed, the weights are 0.05, 0.21, 0.03, 0.54, and 0.17 — three components carry meaningful probability mass. The model is genuinely uncertain, and the five panels visualise that uncertainty.”**(pause 2 s)**“One seed → five imagined next-latents, each weighted by  $\pi_k$ . CMA-ES averages these in dream rollouts to score policies — no real environment touched.”*

The five-futures figure is the focal point — use it. “Five possible futures, weighted” — quotable, pause around it. → Slide 10.

## Slide 10 · 23

12:30 – 13:30 · 1 min

*C · Controller + CMA-ES*

Goal: Reveal the 3,531 parameters for the third time, with the fast-policy GIF on the right.

*“Third component, C — the controller. And this is where the 3,531-parameter story closes.”**[point at the architecture on the left]**“Pure linear: input is  $\text{concat}(z, h)$  — 320 dimensions. Output 11 dimensions: override, direction, accel, decel. **W matrix 320×11 plus bias 11 = 3,531 parameters total.** No hidden layer.”**[point at the CMA-ES Training panel]**“Training is the interesting part. CMA-ES — gradient-free evolution. Popula-*



*tion of 64 candidate weight vectors per generation. Each candidate evaluated by a **dream rollout** — 50 steps, cumulative  $\hat{r}$  becomes the fitness. 290+ generations, about 8 hours on CPU.”*

*[point at the fast-policy GIF on the right]*

*“On the right, the **deployed controller running in real time**. One forward pass per frame:  $V \rightarrow C \rightarrow \text{action}$ . The controller is override-aware — when  $\text{override} = 0$ , it defers to  $A^*$ ; when  $\text{override} = 1$ , it picks an explicit direction. A hybrid by design.”*

“3,531 parameters” — third mention, fully justified. The fast-policy GIF is looping — let it play.  
→ “Now the environment.”

## §4 Environment & Pipeline

3 min · 14:00 – 17:00

Slide 11 · 23

13:30 – 14:30 · 1 min

*RAWSim-O*

Goal: Justify the simulator. Four reason cards.

*“Environment: **RAWSim-O** — open-source warehouse simulator. Four reasons.”*

*[point at each of the four cards]*

*“Reason 1 — Physics-grade fidelity. Real motion model: continuous acceleration, deceleration, turning radius. Not a grid abstraction.”*

*“Reason 2 — On-board camera support. GymServer streams  $96 \times 96 \times 3$  RGB from each robot’s perspective — **identical to what a real robot’s camera sees**. This is critical for my methodology: I train on pixels, not coordinates.”*

*“Reason 3 — Standard MAPF baselines. CBS, BCP, FAR, WHCA\*, OD\_ID, PAS — all built in for honest benchmarking.”*

*“Reason 4 — Open-source and reproducible. C# implementation, standard benchmark instances.”*

*(pause 2s)*

*“RAWSim-O fills the gap between toy gym envs and real warehouses — high fidelity AND visual. Coordinate-only training does not transfer; **pixel-trained world models do**.”*

Four reasons with clear rhythm. “This is critical” — lean in. → Slide 12.

Slide 12 · 23

14:30 – 15:30 · 1 min

*Pipeline*

Goal: End-to-end flow. Two phases visible side-by-side.

*“Two phases.”*

*[point at the offline-phase panel on the left]*

*“**Phase A — Offline Training**. RAWSim-O simulator runs on port 7654. Python client connects via TCP through PettingZoo’s ParallelEnv. Dataset:*



*20 episodes  $\times$  32 bots  $\times$  1,000 frames. From there, train  $V$  (VAE) and  $M$  (MDN-RNN) offline. Then **CMA-ES** trains  $C$  inside dreams — no simulator. Output: **controller\_best.npy**, 3,531 parameters.”*

*[point at the deployment panel on the right]*

**“Phase B — Deployment.** Per robot:  $\text{observation} \rightarrow V \rightarrow z \rightarrow [z \oplus h] \rightarrow C \rightarrow \text{action}.$ ”

*(pause 2 s)*

**“About 1 millisecond per decision. No comm bottleneck. One robot fails — others continue. Each robot is its own brain.”**

“Each robot is its own brain” — quotable, slow delivery.  $\rightarrow$  Slide 13.

## Slide 13 · 23

15:30 – 16:30 · 1 min

*Dataset*

**Goal:** Concrete numbers. Six stat tiles plus two side panels.

**“Dataset statistics.”**

*[point at the six stat tiles]*

**“20 episodes from jenkinsinstance1. 32 bots per episode. 1,000 steps per episode. 640,000 total transitions. Image 96 $\times$ 96 RGB uint8. Reward range  $[-6, +2]$  with mean about 0.003.”**

*[point at the reward-signal panel]*

**“Reward signal breakdown: throughput component is  $+0$  to  $+2$  per step — items handled, orders completed. Collision penalty is  $-1$  to  $-6$  per step. Mean reward of 0.003 means the signal is very sparse.”**

*[point at the action-space panel]*

**“Action space — three integer dimensions: direction\_idx (4 cardinal directions plus override), accel\_mode (3 levels), decel\_mode (3 levels).”**

*(pause 2 s)*

**“Sparse reward is hard. This is one of the reasons I chose **CMA-ES** — gradient-free, robust to sparse signals.”**

Numbers read briskly. “Sparse reward is hard” — owning the problem.  $\rightarrow$  “Now the demos.”

## §5 World-Model Demos

6 min · 17:00 – 23:00

### Slide 14 · 23

16:30 – 18:00 · 1.5 min

*Demo · Ground Truth vs Dream*

**Goal:** First payoff. The GIF on the left, observation list on the right.

*[GIF auto-loops — let it play once before speaking]*

**“First demo. Left side: ground truth from the simulator. Right side: the world model imagining the same future, given the same actions.”**

*(pause 2 s)*

*“The robot has never seen this future. It is hallucinated.”*

*[point at the observation list on the right]*

*“What to watch for: at  $t = 1$ , near-identical — only VAE recon error. At  $t = 10$ – $20$ , the dream tracks scene structure — corridor, sky. At  $t = 30$  onward, the dream loses fine detail; pods blur.”*

*(pause 2 s)*

*“Honest framing: 50-step horizon is challenging — the model was trained for 1-step prediction, errors compound. The model **imagines the future** — **works at short horizon, degrades gracefully**.”*

Let the GIF loop fully before starting. “The robot has never seen this future” — slow, deliberate. → Slide 15.

## Slide 15 · 23

18:00 – 19:30 · 1.5 min

Demo · Counterfactual

Goal: Action conditioning works. Same start, four action policies, four divergent panels.

*[2×2 GIF auto-loops]*

*“Same seed frame, four different action sequences. All four panels start from the same real state.”*

*[point at each panel]*

*“RECORDED: the actual on-policy rollout. ALL-N: every action “forward through corridor” — corridor stays open. ALL-STAY: scene barely changes — bot doesn’t move. ALL-E: scene shows a wall approach — turning right from this position leads into a wall.”*

*(pause 2 s)*

*“The divergence is not noise — it is **systematic**. Different intentions yield different imagined outcomes. Empirical evidence that action conditioning works.”*

*(pause 2 s)*

*“Why this matters: for planning in dreams, the model has to respond differently to different actions. This visual confirms planning is possible.”*

Point at each panel as you name it. “Not noise” — emphasize, slow. → Slide 16.

## Slide 16 · 23

19:30 – 20:30 · 1 min

Demo · Fast Policy

Goal: The deployed controller in real time.

*[fast-policy GIF auto-loops]*

*“The **deployed controller running in real time**. One forward pass per frame:  $V \rightarrow C \rightarrow \text{action}$ . About 1 millisecond per decision.”*

*[point at the overlay reading panel]*

*“Reading the overlay: the arrow is the controller’s chosen direction this frame.”*

*RECORDED is what AgentAStar would have done at the same step. When the controller deviates from  $A^*$ , the arrow changes colour — that is the override.”*

*[point at the pills]*

*“Three properties: **~1 ms per frame**, **no central computation**, **hybrid override-aware**.”*

*(pause 2 s)*

*“That’s fast mode. The next demo — the most insightful one in the talk — is **world thinking**: how CMA-ES uses the world model for training.”*

Faster pace than other demos — set up the climax. “World thinking” — lean into the transition.  
→ Slide 17.

### Slide 17 · 23

20:30 – 22:30 · 2 min

*Climax · The World Model as the CMA-ES Evaluator*

**Goal:** The big reveal. Animated action\_sweep.gif on the left, static summary.png on the right, explanation paragraph and story-rich outcome panel below.

*[let the slide settle for 5 seconds before speaking — animated sweep on the left, static figure on the right]*

*“When training the controller, **CMA-ES never touches the simulator**. The question is: how does it know which candidate weight vector is good?”*

*(pause 3 s)*

*“The answer is on this slide.”*

*[point at the GIF on the left]*

*“On the left is the live loop. Five different action policies — each imagined in the dream for 50 steps. STAY, NORTH, SOUTH, EAST, WEST — all from the same seed. Watch the dream evolve frame by frame for each one.”*

*[point at the static figure on the right]*

*“On the right is the static analysis. Top: the seed plus the five final dream frames. Below: the cumulative reward curve and total bars.”*

*[point at the explanation paragraph and the story-rich outcome panel]*

*“Reading the values from the panel: STAY:  $-20$  — catastrophic, collisions accumulate. WEST:  $+0.14$  — open path, the best of the five. EAST:  $-9$  — wall approach. FAST: the deployed controller chose EAST — good but not optimal.”*

*(pause 2 s)*

*“There is a **disagreement**. The fast policy chose EAST. Imagination says WEST is optimal. This is not a bug — it is the trade-off Ha and Schmidhuber describe: the fast deployed policy trades optimality for speed.”*

*(pause 2 s)*

*“What you are seeing is the **training mechanism made visible**. From a single seed, 5 candidate actions yield 5 different cumulative rewards — visible only through the world model.”*

*(pause 2 s)*

*“This is why CMA-ES training is possible — the world model provides the*

*fitness signal that would otherwise need a real environment.”*

Longest slide. Don't rush. **This is the climax.** Long pause before the reveal. Point at the GIF, then the static figure, then the panel — three discrete reads. “Training mechanism made visible” — quotable. → Slide 18.

## Slide 18 · 23

22:30 – 24:00 · 1.5 min

Demo · CMA-ES Evolution

Goal: Show training over time. PCA trajectory + fitness curve + dream inset.

[evolution GIF auto-loops]

*“Last demo for the world model. 30 saved checkpoints from training, projected to 2D via PCA, animated.”*

[point through the panels]

*“Each dot is the controller at one generation. Colour encodes fitness. From a random init at fitness  $-0.86$ , CMA-ES dives toward the peak at  $+7.05$  — generations 100 to 140 — then drifts past it back to around  $+5$ .”*

[point at the annotated insights panel]

*“Three insights. **PC1 explains 57.7% of parameter variance.** Trajectory is **curved** — CMA-ES is evolutionary, not gradient. And critically: **best fitness is mid-training, not at the end** — that is why I use early stopping, deploying `controller_best.npy`, not the latest checkpoint.”*

Three insights spoken with rhythm. “Best fitness is mid-training, not at the end” — emphasize. → “Now the quantitative side.”

## §6 Benchmark & Results

3 min · 24:00 – 27:00

## Slide 19 · 23

24:00 – 25:00 · 1 min

Quantitative World-Model Quality

Goal: Honest evaluation of imagination fidelity.

[MSE/SSIM plot is shown]

*“Quantitative evaluation. Two metrics: pixel-level MSE on the left, structural similarity SSIM on the right. Three temperatures,  $\pm 1\sigma$  band across two episodes.”*

[point at the four stat tiles on the right]

*“**MSE at  $t=1$ : 0.025** — that's the VAE recon floor. **MSE at  $t=49$ : 0.10** — acceptable for demos. **SSIM at  $t=1$ : 0.68**. **SSIM at  $t=49$ : 0.45** — degraded.”*

[point at the bullet text]

*“ $\tau = 0.5$  — the balanced temperature — is the most stable.  $\tau = 1.0$  is noisier toward the end.  $\tau = 0$  is over-confident.”*

(pause 2 s)

*“**Reliable horizon  $\approx 30$  steps** — matches my CMA-ES rollout config of 50 total steps.”*

Numbers read briskly. → Slide 20.

## Slide 20 · 23

25:00 – 26:30 · 1.5 min

### Centralized PP Comparison

**Goal:** Set up what I need to beat. Throughput chart on left, comparison table on right.

[throughput chart shown]

**“Benchmark from RAWSim-O: 7 path-planning algorithms, 3 random seeds, jenkinsinstance1, 32 bots.”**

[point at the table]

**“The centralized planners — top six rows, FAR, CBS, BCP, WHCAvStar, WHCAnStar, OD\_ID — achieve about 3,150 items handled per episode with zero collisions.”**

[point at the highlighted last row]

**“AgentAStar — decentralized rule-based — handles slightly more items, 3,197, but with 1,339 collisions per episode.”**

(pause 2 s)

**“The classic trade-off: centralized = clean coordination but fragile. Decentralized rule-based = collisions explode.”**

(pause 2 s)

**“The bar to clear: minimise collisions while staying decentralized.”**

“1,339 collisions” — emphasize. The research question is implicit — let the table do the work. → Slide 21.

## Slide 21 · 23

26:30 – 27:30 · 1 min

### Where We Stand

**Goal:** Honest about state. Radar chart + status table + summary panel.

[radar chart and tables shown]

**“Status, transparently.”**

[point at the radar]

**“Centralized planners across multi-metric — throughput, distance, energy, kinematics — converge to a similar profile.”**

[point at the status table]

**“Three rows. Centralized planners: not decentralized, zero collisions, mature baselines. AgentAStar: decentralized, but 1,339+ collisions. Our world model, in-dream: decentralized, collisions TBD, status training done · real-eval next.”**

[point at the in-dream summary panel]

**“In-dream summary: trained on AgentAStar baseline data; best controller at gen 100–140 with  $\Sigma$  predicted reward +7.05. Real-environment evaluation pending.”**

(pause 2 s)

*“The architecture is set; we now need integration testing. The deploy bridge has been built — what remains is just running the 30-episode benchmark.”*

“Transparently” — sets the tone. “Bridge built — just running the benchmark” — pre-empts the obvious advisor question. → Slide 22.

## §7 Remaining Work & Close

1.5 min · 27:00 – 28:30

### Slide 22 · 23

27:30 – 28:00 · 30 sec

#### Remaining Tasks

**Goal:** The deploy bridge is done — only two tasks remain.

*[two-card layout shown]*

*“Two concrete tasks. The deploy bridge is already built — that is why it is not on this list.”*

*[point at Task 01]*

*“Task 01 — Hyperparameter sweep.  $\beta$ -KL weight for the VAE, latent dim across {32, 64, 128, 224}. GMM components {5, 7, 10} for the MDN-RNN. CMA-ES  $\sigma$ -initial, population, dream horizon. Reward shaping — collision penalty weight.”*

*[point at Task 02]*

*“Task 02 — Larger training set. Currently at 100 episodes — 640,000 transitions. Target 100+ episodes with diverse instance configs — jenkinsinstance2, jenkinsinstance3. Retrain V and M, re-run CMA-ES.”*

*(pause 2 s)*

*“Mostly parallelizable tuning and data-scaling work.”*

Brisk and business-like. “Deploy bridge already built” — repeat for the advisor. → Slide 23.

### Slide 23 · 23

28:00 – 28:30 · 30 sec; image holds

#### Closing

**Goal:** Memorable closing. Three sentences, one final caption, then silence.

*[closing slide is shown — three sentences on the left, the Scenario E summary as the final visual on the right]*

*[slow, deliberate, three sentences]*

*“Three sentences.”*

*(pause 2 s)*

*“World models reframe RMFS control as imagination plus a tiny linear policy.”*

*(pause 2 s)*

*“The model learns visual dynamics from pixels alone, with action conditioning that drives different imagined futures.”*

*(pause 2 s)*

*“With 3,531 controller parameters and zero real-environment training steps,*

*we have a decentralized agent ready for real-environment benchmarking.”*

*(pause 3 s)*

*“My closing line:”*

*(pause 2 s)*

*“Each frame is a future the robot considered.”*

Three sentences spoken **slowly, deliberately, clearly**. 2-second pauses between sentences. “Each frame is a future the robot considered” — the final memorable phrase. **Stay silent**. Don’t fill the silence. Let it land. If the advisor stays silent past 5 seconds: “I am happy to take questions.”

## §8 Q&A Preparation

5 min budget

### Q1 · “Why CMA-ES instead of PPO/SAC?”

*“Three reasons. (1) The action space contains a discrete argmax — gradient flow is blocked, gradient-based methods need a continuous relaxation. (2) Reward is very sparse — mean 0.003. (3) CMA-ES is gradient-free with parallel candidate evaluation — fits dream-based training naturally.”*

### Q2 · “Isn’t 20 episodes too small?”

*“Acknowledged on Slides 13 and 22. 20 episodes  $\times$  32 bots  $\times$  1,000 steps = 640,000 transitions — order of magnitude similar to H&S 2018. What matters more is diversity, which is why Task 02 expands to 100+ episodes with multiple instance configs.”*

### Q3 · “How does the controller handle multiple bots?”

*“Each bot runs an independent WorldModelBot instance — no shared state, no communication. That is the definition of decentralized. Collision risk is higher than centralized PP — the long-term mitigation is multi-agent emergent communication, beyond the current scope.”*

### Q4 · “Is the comparison with centralized PP fair?”

*“Setup is identical: jenkinsinstance1, 32 bots, same config. What differs: centralized PP has access to ground-truth state for every bot; the world-model controller only has the egocentric camera. So it is a fundamental handicap — but that is exactly the point: vision-only matches the real warehouse setting.”*

### Q5 · “What’s your original innovation beyond H&S 2018?”

*“Three contributions. (1) 6-channel frame stacking for motion cueing. (2) Override-aware action encoding — hybrid  $A^*$  + learned override. (3) A visualization toolkit — the per-state action sweep on Slide 17 and the training-trajectory PCA on Slide 18 — visualizations not present in any existing world-model paper.”*

### Q6 · “Why a horizon of 50 in CMA-ES rollouts?”

*“A trade-off between fidelity and signal. SSIM at Slide 19 drops from 0.68 to 0.45 over 50 steps. Longer than that, the dream becomes unreliable; shorter, the planning*



*horizon is too narrow. 50 steps is the empirical sweet spot.”*

## Q7 · “Why pixel reconstruction over JEPA?”

*“Two reasons. The pixel-reconstruction lineage is mature — VAE plus MDN-RNN is widely reproduced. And having a decoder gives me interpretability tools — every dream visualization on Slides 14, 15, 17, and 18 requires a decoder. JEPA is on my exploration list.”*

## §9 Response Protocols

### If the advisor stops you mid-talk

1. **Stop immediately** — don’t finish the sentence.
2. **Listen fully.**
3. **Pause 1–2 seconds before answering** — signal thoughtfulness.
4. Answer the question **directly first**, then the related context.
5. If you don’t know: *“I haven’t tested that explicitly, but my hypothesis would be...”*

### If the advisor looks bored

- Skip technical details on less-critical slides.
- Jump to a demo earlier — visuals maintain attention.

### If the advisor looks skeptical

- Don’t be defensive.
- Acknowledge: *“You’re right — that’s a valid concern. I address it in [Slide X].”*

## §10 Memorable Phrases

Sentences the advisor should remember after the talk:

1. **“Each robot is its own brain.”** *(Slide 12)*
2. **“Five possible futures, weighted.”** *(Slide 09)*
3. **“The training mechanism made visible.”** *(Slide 17)*
4. **“Each frame is a future the robot considered.”** *(closing)*
5. **“3,531 parameters and zero real-environment training steps.”** *(Slide 23)*
6. **“The robot has never seen this future.”** *(Slide 14)*

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*End of speaking script.*

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